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Discrete Choice Experiment: Organic and Certified Tomato Valuation

A Cross-Country Comparison of Chinese and
Japanese Consumer Preferences

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1. Introduction and Motivation

Food certification labels such as “organic” and “specially certified” have become increasingly important quality signals in East Asian markets. However, how consumers respond to these labels—and whether external information can shift their valuations—remains an open empirical question, particularly when comparing markets with different food safety histories and cultural attitudes toward certification.

This study addresses the following research question: Do Chinese and Japanese consumers value organic and specially certified tomatoes differently, and does external information about certification affect their willingness to pay? A secondary question concerns preference heterogeneity: are there distinct consumer types who respond fundamentally differently to label information?

To answer these questions, we use data from a Discrete Choice Experiment (DCE) conducted in China and Japan, estimated using the Apollo package in R [2]. Five models are estimated sequentially: a baseline Multinomial Logit (MNL), a country-specific MNL with WTP estimates, an information-treatment interaction MNL, a two-class Latent Class Model (LCM) capturing preference heterogeneity, and an Attribute Non-Attendance (ANA) model as the behavioural extension. The main finding is that unobserved preference heterogeneity—not country differences or information treatments—is the primary driver of variation in label valuations: approximately 75% of respondents are label-sensitive, while 25% largely disregard certification labels.

2. Data Description

2.1 Sample and Experimental Design

The dataset comprises 834 respondents: 422 from China and 412 from Japan. Each respondent completed 8 choice tasks, yielding 6,672 choice observations. Each task presented two labelled tomato products (Product A and Product B) and an opt-out alternative (No-buy), forming a three-alternative design.

Two attributes varied across alternatives. Price was denominated in local currency: RMB/500g for China (range approximately 3–19 RMB) and JPY/500g for Japan (range approximately 170–330 JPY). Label took three levels: ordinary (baseline), organic, or specially certified.

Prior to the choice tasks, respondents were randomly assigned to one of four information treatment groups: EIC (control, no information), EIO (organic information only), EIS (special certification information only), or EIOS (both types of information). Individual-specific consumer segments (SEG1–SEG5, with SEG5 as reference) were pre-coded in the data. The dataset is drawn from the study by Yang [3], which examines the role of brand commitment and external information in urban consumers’ organic produce choices in Japan and China. The anonymised choice-experiment data are publicly available on

Mendeley Data [4] under a CC BY 4.0 licence.

2.2 Descriptive Statistics

Table 1: Choice shares by country and product type

Country	Product type	n	Share (%)
China	Special certification	1,409	41.7
	Organic	1,058	31.3
	Ordinary	554	16.4
	No-buy	355	10.5
Japan	Special certification	1,387	42.1
	Organic	875	26.5
	Ordinary	714	21.7
	No-buy	320	9.7

Table 1 shows that in both countries, specially certified tomatoes command the highest choice share ($\approx 42\%$), followed by organic. Japanese consumers chose ordinary tomatoes more frequently than Chinese consumers (21.7% vs. 16.4%), while no-buy rates are similar ($\approx 10\%$).

Table 2: Consumer segment distribution by country

Segment	China (n)	China (%)	Japan (n)	Japan (%)
SEG1	24	5.7	98	23.8
SEG2	96	22.7	116	28.2
SEG3	41	9.7	118	28.6
SEG4	63	14.9	64	15.5
SEG5 (ref.)	198	46.9	16	3.9
Total	422	100.0	412	100.0

The segment distributions differ substantially between countries (Table 2): Chinese respondents are concentrated in SEG5 (47%), while Japanese respondents are more evenly spread across SEG1–SEG3.

3. Model Specifications

Five models are estimated sequentially, each building on the previous. All models treat ordinary tomato as the baseline label and include an alternative-specific constant (ASC) for the no-buy option. Estimation uses the BGW algorithm [1] in Apollo 0.3.7 [2].

3.1 M1 — Baseline Pooled MNL

The baseline model pools all respondents with a single price coefficient and two label dummies. The utility of alternative j for respondent i is:

$$V_{ij} = \beta_{\text{price}} \cdot p_{ij} + \beta_{\text{organic}} \cdot \text{org}_{ij} + \beta_{\text{spec}} \cdot \text{spec}_{ij} \quad (1)$$

$$V_{i,\text{nobuy}} = \text{ASC}_{\text{nobuy}} \quad (2)$$

This model establishes whether labels matter at all, but ignores country differences and conflates RMB and JPY price units.

3.2 M2 — Country-Specific MNL with Segment Effects

M2 separates price and label coefficients by country and adds consumer segments to the no-buy utility as individual-specific variables (SEG5 = reference):

$$V_{ij} = \beta_{\text{price}}^{\text{cn}} \cdot p_{ij}^{\text{cn}} + \beta_{\text{price}}^{\text{jp}} \cdot p_{ij}^{\text{jp}} + (\beta_{\text{org}}^{\text{cn}} \cdot \text{CN}_i + \beta_{\text{org}}^{\text{jp}} \cdot \text{JP}_i) \text{org}_{ij} + (\beta_{\text{spec}}^{\text{cn}} \cdot \text{CN}_i + \beta_{\text{spec}}^{\text{jp}} \cdot \text{JP}_i) \text{spec}_{ij}$$

$$V_{i,\text{nobuy}} = \text{ASC}_{\text{nobuy}}^{\text{cn}} \cdot \text{CN}_i + \text{ASC}_{\text{nobuy}}^{\text{jp}} \cdot \text{JP}_i + \sum_{s=1}^4 \gamma_s \cdot \text{SEG}_{is} \quad (3)$$

Willingness to pay is derived as $\text{WTP}_\ell = -\beta_\ell / \beta_{\text{price}}$.

3.3 M3 — Information-Treatment Interaction MNL

M3 extends M2 by adding interactions between label dummies and information treatment indicators:

$$\text{org_effect}_i = \beta_{\text{org}} + \beta_{\text{org,eio}} \cdot \text{EIO}_i + \beta_{\text{org,eios}} \cdot \text{EIOS}_i \quad (4)$$

$$\text{spec_effect}_i = \beta_{\text{spec}} + \beta_{\text{spec,eis}} \cdot \text{EIS}_i + \beta_{\text{spec,eios}} \cdot \text{EIOS}_i \quad (5)$$

The control group (EIC) is the reference; positive interactions imply that information raises label valuation.

3.4 M4 — Two-Class Latent Class Model

M4 allows respondents to belong to one of two latent classes, each with its own preference parameters. Class membership is modelled as:

$$V_i^{\text{class2}} = \delta_{\text{class2}} + \gamma_{\text{china}} \cdot \text{CN}_i + \sum_{s=1}^4 \gamma_{s,\text{class2}} \cdot \text{SEG}_{is} \quad (6)$$

where Class 1 is the reference (utility = 0). Class shares (π_1, π_2) are estimated via a logistic allocation function. This model directly captures unobserved preference heterogeneity.

3.5 M5 — Attribute Non-Attendance Extension (Behavioural)

M5 implements a two-class ANA model: Class 1 (label-attending) considers all attributes, while Class 2 (label-non-attending) responds to price only. Price coefficients are shared across classes for identification. Class membership depends on information treatment assignment:

$$V_i^{\text{attend}} = \delta_{\text{att}} + \gamma_{\text{eio}} \cdot \text{EIO}_i + \gamma_{\text{eis}} \cdot \text{EIS}_i + \gamma_{\text{eios}} \cdot \text{EOS}_i \quad (7)$$

where no-attend is the reference class. This specification directly tests whether information treatments shift the probability of label attendance.

4. Results and Interpretation

4.1 Specification Exploration: Number of Latent Classes

To justify the choice of two latent classes in M4, a three-class LCM (M4b) was also estimated with the same class membership covariates (country and consumer segments). Table 3 summarises the fit statistics across specifications; the one-class benchmark corresponds to M2 (country-specific MNL).

Table 3: Latent class specification exploration

Specification	LL	k	AIC	BIC	ρ^2
1-class (M2 Country MNL)	−5,023.04	12	10,070.08	10,151.74	0.315
2-class LCM (M4)	−4,370.19	16	8,772.38	8,881.27	0.404
3-class LCM (M4b)	−3,729.63	27	7,513.25	7,697.01	0.491

Note: ρ^2 computed against equal-shares $\text{LL}(0) = -7,329.94$. The three-class model uses 11 additional parameters relative to the two-class model.

Although the three-class model achieves lower AIC and BIC, inspection of the estimated parameters reveals two identification problems that undermine its substantive interpretation. First, the China price coefficient for Class 1 carries a positive sign ($\hat{\beta}_{\text{price,cn}}^{(1)} = +0.145$, robust $t = +3.66$), implying that Chinese consumers in that class prefer higher-priced tomatoes — an implausible result inconsistent with standard consumer theory. Second, the Japan price coefficient for Class 3 is statistically indistinguishable from zero ($\hat{\beta}_{\text{price,jp}}^{(3)} = -0.002$, robust $t = -0.64$), making WTP ratios undefined for that group. These anomalies suggest that, despite its better information criteria, the three-class solution is behaviourally less reliable and less suitable for WTP interpretation than the two-class model.

Because likelihood-ratio testing is not straightforward for latent class models — the standard asymptotic chi-squared approximation does not hold when comparing models with different numbers of classes — model selection relies primarily on AIC, BIC, parameter interpretability, and behavioural plausibility. On these grounds, the two-class LCM (M4) is retained as the preferred heterogeneity specification: it provides a large improvement in fit over the one-class benchmark, all price coefficients carry the correct sign, and the two

classes have a clear and economically meaningful interpretation.

4.2 Model Fit Comparison

Table 4: Model fit statistics

Model	LL	k	AIC	BIC	ρ^2
M1: Pooled MNL	-5,833.83	4	11,675.65	11,702.88	0.204
M2: Country MNL + SEG	-5,023.04	12	10,070.08	10,151.74	0.315
M3: Info Interactions	-5,010.41	14	10,048.81	10,144.09	0.316
M4: 2-class LCM	-4,370.19	16	8,772.38	8,881.27	0.404
M5: ANA 2-class	-4,609.82	10	9,239.63	9,307.69	0.371

Note: $\rho^2 = 1 - \text{LL}(\text{final})/\text{LL}(0)$, where $\text{LL}(0) = -7,329.94$ (equal shares). k = number of estimated parameters. M4 (shaded) is the best-fitting model by both AIC and BIC. LR tests: M2 vs M1: $\Delta\text{LL} = 810.8$, $\Delta k = 8$, $p < 0.001$; M4 vs M2: $\Delta\text{LL} = 652.9$, $\Delta k = 4$, $p < 0.001$.

Table 4 shows a clear progression in model fit. The country split (M2) yields a large improvement over the pooled baseline ($\text{LR} = 1,621.6$, $p < 0.001$), confirming that pooling China and Japan is inappropriate. Information-treatment interactions (M3) add only marginal improvement ($\text{LR} = 25.3$, $p < 0.001$). The latent class model (M4) delivers the largest gain ($\text{LR} = 1,305.7$ vs. M2, $p < 0.001$). Among the five main models, M4 provides the best fit by both AIC and BIC. Although the exploratory three-class LCM (M4b) achieves lower information criteria, it is not retained as the preferred specification because of implausible and weakly identified price coefficients, as discussed in Section 4.1.

4.3 Baseline Results (M1)

All M1 coefficients are highly significant and correctly signed: $\hat{\beta}_{\text{price}} = -0.006$ ($t = -18.1$), $\hat{\beta}_{\text{organic}} = 0.231$ ($t = 5.95$), $\hat{\beta}_{\text{spec}} = 0.943$ ($t = 19.15$), and $\text{ASC}_{\text{nobuy}} = -2.005$ ($t = -25.8$). Special certification commands a substantially larger utility premium than organic, and the strong negative no-buy ASC reflects a baseline preference for purchasing over opting out. These results establish that certification labels matter significantly.

4.4 Country Differences and Willingness to Pay (M2)

Table 5: Willingness to pay — M2 country-specific MNL

Country	Label attribute	WTP	Currency
China	Organic vs Ordinary	+8.34	RMB/500g
	Special cert. vs Ordinary	+6.51	RMB/500g
Japan	Organic vs Ordinary	+40.19	JPY/500g
	Special cert. vs Ordinary	+37.64	JPY/500g

Chinese consumers show higher WTP for organic (+8.34 RMB) than for special certification (+6.51 RMB), whereas Japanese consumers value both labels similarly (+40.19

vs. +37.64 JPY). Point estimates suggest these differences are economically meaningful, although formal inference would require delta-method or bootstrap confidence intervals. A particularly striking finding is Japan’s much stronger aversion to the no-buy option ($ASC_{\text{nobuy}}^{\text{jp}} = -6.69$ vs. -2.38 for China), indicating that Japanese respondents have a substantially lower baseline propensity to choose the no-buy alternative, conditional on the observed attribute levels in the experiment. Among consumer segments, SEG4 shows a weakly lower propensity to opt out ($\hat{\gamma}_{\text{seg4}} = -0.459$, robust $t = -1.62$), though the effect is only marginally significant at conventional levels.

4.5 Information Treatment Effects (M3)

Table 6: Information-adjusted WTP — M3

Attribute	Scenario	WTP China (RMB)	WTP Japan (JPY)
Organic	Control (EIC)	6.57	41.3
	EIO (organic info)*	9.71	61.0
	EIOS (both info)	7.60	47.8
Special cert.	Control (EIC)	5.97	37.5
	EIS (special info)	6.61	41.5
	EIOS (both info)	7.16	45.0

Note: * Only `b_org_eio` is statistically significant ($\hat{\beta} = 0.454$, robust $t = 3.39$). All other interaction terms have robust $|t| < 1.1$.

Only receiving organic-only information (EIO) significantly raises WTP for the organic label, from 6.57 to 9.71 RMB in China and from 41.3 to 61.0 JPY in Japan (Table 6). All other information interactions are statistically insignificant, suggesting that targeted organic information can shift preferences but combined or certification-only messages have no detectable effect on WTP.

4.6 Preference Heterogeneity — Latent Class Model (M4)

M4 is the best-fitting model and reveals the central finding: respondents fall into two qualitatively distinct latent classes.

Class 1 (label-sensitive, 74.8%) exhibits strong, highly significant label preferences: $\hat{\beta}_{\text{organic}}^{(1)} = 1.391$ (robust $t = 16.3$) and $\hat{\beta}_{\text{spec}}^{(1)} = 1.219$ (robust $t = 16.3$). The strong no-buy aversion ($ASC_{\text{nobuy}}^{(1)} = -9.54$) reflects near-universal purchasing intent.

Class 2 (label-indifferent, 25.2%) treats labels as essentially irrelevant: $\hat{\beta}_{\text{organic}}^{(2)} = -0.235$ (robust $t = -1.26$) and $\hat{\beta}_{\text{spec}}^{(2)} = 0.120$ (robust $t = 0.84$) are both statistically indistinguishable from zero.

Chinese respondents are significantly more likely to belong to Class 2 ($\hat{\gamma}_{\text{china}} = 0.676$, robust $t = 3.12$). Posterior class assignment confirms this: 29.6% of Chinese respondents

Table 7: Selected M4 parameter estimates by latent class

Parameter	Class 1 (74.8%)		Class 2 (25.2%)	
	Est.	Rob. t	Est.	Rob. t
$\beta_{\text{price,cn}}$	-0.121	-10.53	-0.266	-8.38
$\beta_{\text{price,jp}}$	-0.026	-12.14	-0.014	-13.64
β_{organic}	+1.391	+16.26	-0.235	-1.26
β_{spec}	+1.219	+16.34	+0.120	+0.84
$\text{ASC}_{\text{nobuy}}$	-9.539	-13.05	-2.805	-10.06
δ_{class2} (intercept)	—		-1.539	-6.34
γ_{china}	—		+0.676	+3.12

Note: Hessian is negative definite (max eigenvalue = -4.02). Class 1 is the reference class. γ_{china} = effect of Chinese nationality on the probability of belonging to Class 2. Segment effects ($\gamma_{\text{seg1-4}}$) are all non-significant and omitted.

(125/422) are assigned to Class 2, versus only 19.9% of Japanese respondents (82/412).

Table 8: Class-specific WTP — M4 latent class model

Class	Country	Organic WTP	Special cert. WTP
Class 1 (label-sensitive)	China	+11.46 RMB	+10.05 RMB
	Japan	+53.83 JPY	+47.21 JPY
Class 2 (label-indifferent)	China	-0.88 RMB (n.s.)	+0.45 RMB (n.s.)
	Japan	-16.26 JPY (n.s.)	+8.29 JPY (n.s.)

Note: n.s. = not statistically significant. Class 2 label coefficients have robust $|t| < 1.3$. WTP for Class 2 should not be interpreted as economically meaningful.

For label-sensitive consumers (Class 1), WTP estimates substantially exceed the pooled M2 estimates (Table 8), confirming that pooled WTP understates the true preferences of the majority of consumers.

4.7 Attribute Non-Attendance Extension (M5)

The ANA model constitutes the behavioural extension of this analysis. Importantly, it is implemented as a latent-class structure — respondents are probabilistically assigned to a label-attending or label-non-attending class — thereby combining the behavioural extension with preference heterogeneity, as required by the course specification. The attending class considers all attributes (price, organic, special certification), while the non-attending class responds to price only; price coefficients are shared across both classes for identification.

The model identifies 76.2% of respondents as label-attending, broadly consistent with M4's

Class 1 share (74.8%). The baseline log-odds of attendance ($\hat{\delta}_{\text{att}} = 1.270$, robust $t = 7.03$) is large and significant. Among attending consumers, label utilities are precisely estimated: $\hat{\beta}_{\text{organic,att}} = 1.260$ (robust $t = 20.4$) and $\hat{\beta}_{\text{spec,att}} = 1.229$ (robust $t = 19.2$).

Critically, all three information-treatment effects on attendance probability are statistically insignificant: $\hat{\gamma}_{\text{eio}} = 0.122$ ($t = 0.48$), $\hat{\gamma}_{\text{eis}} = -0.404$ ($t = -1.70$), $\hat{\gamma}_{\text{eios}} = -0.103$ ($t = -0.42$). Information treatments do not shift whether consumers attend to label attributes; they affect only the valuations of attended labels (as seen in M3 for organic-only information), not the cognitive decision to consider labels.

Table 9: Average posterior attendance probability by country and information group — M5

Country	Information group	Avg. attendance prob.	n
China	EIC (Control)	0.726	103
	EIO (Organic info)	0.732	107
	EIOS (Both info)	0.661	104
	EIS (Special info)	0.637	108
Japan	EIC (Control)	0.833	109
	EIO (Organic info)	0.875	99
	EIOS (Both info)	0.870	99
	EIS (Special info)	0.773	105

Note: Cross-group differences within each country are not statistically significant (all $\hat{\gamma}$ have robust $|t| < 1.7$).

Table 9 shows that Japanese consumers consistently exhibit higher label attendance probabilities (0.77–0.88) than Chinese consumers (0.64–0.73), consistent with M4’s finding that Chinese respondents are more likely to be label-indifferent. This cross-country pattern in attendance probabilities provides further behavioural evidence complementing the latent class results.

5. Conclusions and Key Findings

This study examined consumer valuation of organic and specially certified tomatoes in China and Japan using a DCE and six estimated models (including a three-class specification for exploration). Four key findings emerge.

Finding 1 — Both countries strongly prefer certified tomatoes. Special certification commands the highest choice share in both China (41.7%) and Japan (42.1%), with organic second. Opt-out rates are low ($\approx 10\%$), indicating a high baseline propensity to choose one of the tomato products in the experimental setting.

Finding 2 — Country differences are substantial but asymmetric. Chinese consumers show higher WTP for organic (8.34 RMB) than for special certification (6.51 RMB), while

Japanese consumers value both labels similarly (≈ 40 and ≈ 38 JPY). Japanese respondents display an exceptionally strong aversion to the no-buy option ($ASC_{\text{nobuy}}^{\text{JP}} = -6.69$).

Finding 3 — Preference heterogeneity is the dominant source of variation. The two-class LCM (M4) is retained as the preferred model because it balances fit, parsimony, and behavioural interpretability: approximately 75% of respondents are label-sensitive consumers willing to pay substantial premiums, while 25% are label-indifferent. Chinese respondents are significantly more likely to be label-indifferent. Ignoring this heterogeneity (as in M1–M3) produces misleading average WTP estimates that mask the bimodal distribution of preferences.

Finding 4 — Information treatments have limited and specific effects. Only organic-focused information (EIO) significantly raises WTP for the organic label; other treatments have no detectable effect on WTP (M3). More fundamentally, no information treatment shifts the probability of label attendance (M5): consumers who attend to labels do so regardless of information received. This suggests that certification campaigns may need to target label-indifferent consumers more directly, potentially through channels beyond pre-choice information provision.

Limitations. WTP confidence intervals were not computed (delta method or bootstrapping would be needed for formal inference on WTP differences). The segment variables (SEG1–SEG5) were not significant predictors of latent class membership, suggesting that the observed segmentation does not capture the underlying behavioural heterogeneity revealed by M4.

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